

Core identity for the general- d product-density lemma

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Abstract

Step 0 of the general- d equal-ratio monomial milestone (Astra #12 (c)). The pushforward of $\prod_i t_i^{\lambda-1} dt$ on $(0, 1]^d$ under $t \mapsto \prod_i t_i$ has density $z^{\lambda-1}(-\log z)^{d-1}/(d-1)!$; Astra identified the one-dimensional identity that drives the induction, $\int_z^1 (-\log(z/t))^{d-1}/t dt = (-\log z)^d/d$. It is proved here in the form $\int_z^1 (\log t - \log z)^d/t dt = (-\log z)^{d+1}/(d+1)$ for $0 < z \leq 1$ (`integral_log_sub_pow_div`) by the fundamental theorem of calculus with antiderivative $(\log t - \log z)^{d+1}/(d+1)$. Zero `sorry/axiom`.

1 The things formalised

```
theorem hasDerivAt_log_sub_pow_div (z t) (ht : 0 < t) (d : ℕ) :
  HasDerivAt (fun t => (log t - log z)^(d+1) / (d + 1)) ((log t - log z)^d / t) t
theorem integral_log_sub_pow_div (z) (hz : 0 < z) (hz1 : z < 1) (d : ℕ) :
  ∫ t in z..1, (log t - log z)^d / t = (-log z)^(d+1) / (d + 1)
```

2 Lean notes

- `intervalIntegral.integral_eq_sub_of_hasDerivAt` needs the derivative on `uIcc z 1` (`uIcc_of_le`) and interval integrability, supplied by `ContinuousOn.intervalIntegrable` for the quotient of continuous functions with nonvanishing denominator on $[z, 1]$.
- `HasDerivAt.congr_deriv` with `field_simp` matches the derivative of the antiderivative to the integrand; `volume` needs `open MeasureTheory`.
- The evaluation at the endpoints is `simp [Real.log_one]` ($0^{d+1} = 0$ at $t = z$).